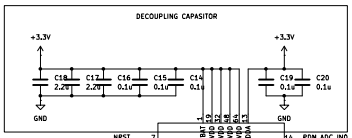
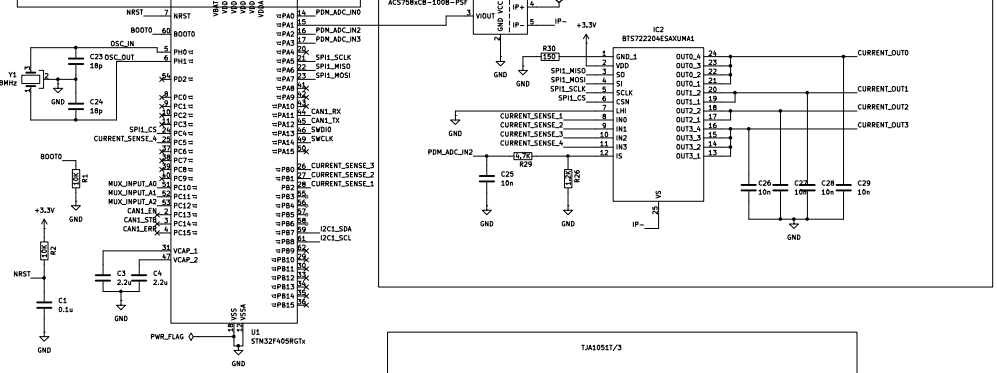
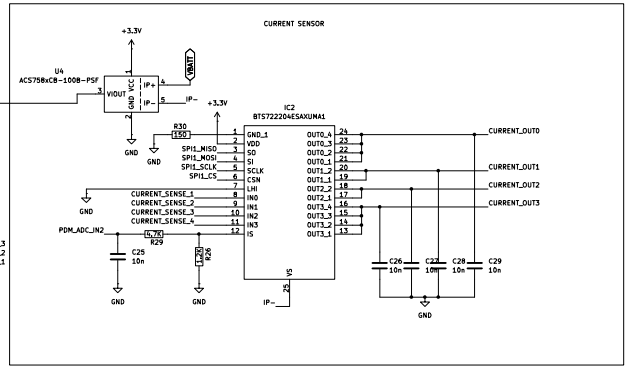
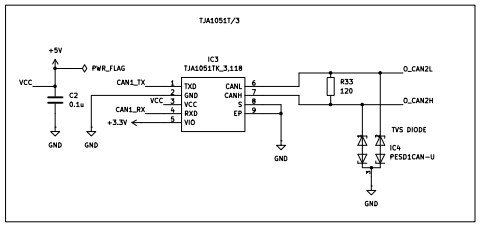
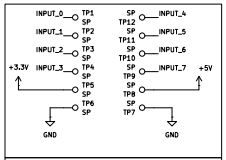

POWER
VBAT SENSING
VBATT
R24
39K
U5A
LM358_DFN
PDM_ADC_IN0
R25
4.7K
GND
3.3V
+LDIN
PWR_FLAG
U6
AMS1117-3.3
VI
VO
GND
+3.3V
C21
10u
C22
0.1u
0.1u
C34
10u
C41
TP13
TestPoint
VBATT
D3
PMEG3020EJ
C38
10u
C42
10u
C43
10u
U7
AP62250WU-7
VIN
BST
SW
EN
FB
GND
L2
4.7uH
C44
330n
+5V
R27
C45
22u
C46
22u
D2
POWER_LED
R28
100K
R30
150
D5
PMEG3020EJ
+LDIN
VBATT
P
N
SHIELD
J3
XT60PW-M
LI-PO 3S-4S TO 5V

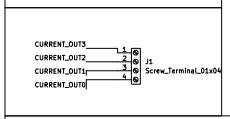

DECOUPLING CAPASITOR
+3.3V
C18
2.2u
C17
2.2u
C16
0.1u
C15
0.1u
C14
0.1u
C19
0.1u
C20
0.1u
GND

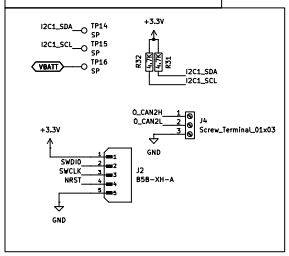

NRST
BOOT0
OSC_IN
OSC_OUT
Y1
8MHz
C23
18p
C24
18p
R1
100
R2
10K
C1
0.1u
C3
2.2u
C4
2.2u
VCAP_1
VCAP_2
PWR_FLAG
U1
STM32F405RGTx
VSS
VSSA
SPI1_CS
SPI1_SCLK
SPI1_MISO
SPI1_MOSI
CAN1_RX
CAN1_TX
SWDIO
SWCLK
CURRENT_SENSE_4
CURRENT_SENSE_3
CURRENT_SENSE_2
CURRENT_SENSE_1
MUX_INPUT_A0
MUX_INPUT_A1
MUX_INPUT_A2
CAN1_EN
CAN1_STB
CAN1_ERR
I2C1_SDA
I2C1_SCL
PDM_ADC_IN0
PDM_ADC_IN2
PDM_ADC_IN3

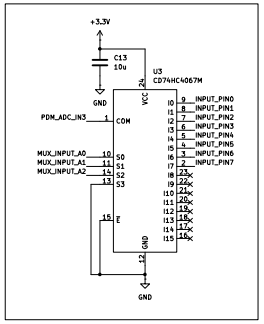

CURRENT SENSOR
U4
ACS758xCB-100B-PSF
VIOUT
IP+
IP-
VCC
GND
VBATT
+3.3V
R30
150
IC2
BTS722204ESAXUMA1
GND_1
VDD
SO
SI
SCLK
CSN
LHI
IN0
IN1
IN2
IN3
IS
OUT0_4
OUT0_3
OUT0_2
OUT0_1
OUT1_2
OUT1_1
OUT2_2
OUT2_1
OUT3_4
OUT3_3
OUT3_2
OUT3_1
SPI1_MISO
SPI1_MOSI
SPI1_SCLK
SPI1_CS
CURRENT_SENSE_1
CURRENT_SENSE_2
CURRENT_SENSE_3
CURRENT_SENSE_4
PDM_ADC_IN2
R29
4.7K
R26
1.2K
C25
10n
C26
10n
C27
10n
C28
10n
C29
10n
CURRENT_OUT0
CURRENT_OUT1
CURRENT_OUT2
CURRENT_OUT3

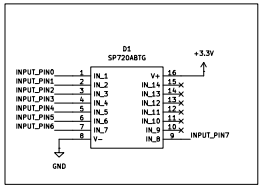

TJA1051T/3
IC3
TJA1051TK_3.118
+5V
PWR_FLAG
VCC
C2
0.1u
CAN1_TX
CAN1_RX
TXD
GND
VCC
RXD
VIO
CANL
CANH
S
EP
R33
120
O_CAN2L
O_CAN2H
TVS DIODE
IC4
PESD1CAN-U

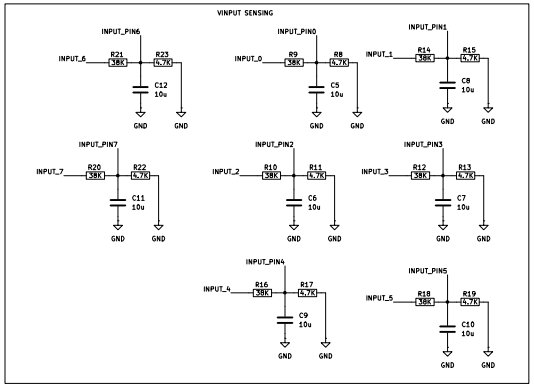

INPUT_0
INPUT_1
INPUT_2
INPUT_3
INPUT_4
INPUT_5
INPUT_6
INPUT_7
TP1
TP2
TP3
TP4
TP5
TP6
TP7
TP8
TP9
TP10
TP11
TP12
SP
+3.3V
+5V
GND

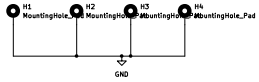

CURRENT_OUT3
CURRENT_OUT2
CURRENT_OUT1
CURRENT_OUT0
J1
Screw_Terminal_01x04


I2C1_SDA
I2C1_SCL
TP14
TP15
TP16
SP
VBATT
+3.3V
R32
4.7K
R31
4.7K
I2C1_SDA
I2C1_SCL
O_CAN2H
O_CAN2L
J4
Screw_Terminal_01x03
SWDIO
SWCLK
NRST
J2
BSB-XH-A
GND


+3.3V
C13
10u
GND
U3
CD74HC4067M
VCC
PDM_ADC_IN3
COM
MUX_INPUT_A0
MUX_INPUT_A1
MUX_INPUT_A2
S0
S1
S2
S3
E
I0
I1
I2
I3
I4
I5
I6
I7
I8
I9
I10
I11
I12
I13
I14
I15
INPUT_PIN0
INPUT_PIN1
INPUT_PIN2
INPUT_PIN3
INPUT_PIN4
INPUT_PIN5
INPUT_PIN6
INPUT_PIN7


D1
SP720ABTG
+3.3V
INPUT_PIN0
INPUT_PIN1
INPUT_PIN2
INPUT_PIN3
INPUT_PIN4
INPUT_PIN5
INPUT_PIN6
INPUT_PIN7
V+
V-
GND


VINPUT SENSING
INPUT_PIN6
INPUT_6
R21
39K
R23
4.7K
C12
10u
INPUT_PIN0
INPUT_0
R9
39K
R8
4.7K
C5
10u
INPUT_PIN1
INPUT_1
R14
39K
R15
4.7K
C8
10u
INPUT_PIN7
INPUT_7
R20
39K
R22
4.7K
C11
10u
INPUT_PIN2
INPUT_2
R10
39K
R11
4.7K
C6
10u
INPUT_PIN3
INPUT_3
R12
39K
R13
4.7K
C7
10u
INPUT_PIN4
INPUT_4
R16
39K
R17
4.7K
C9
10u
INPUT_PIN5
INPUT_5
R18
39K
R19
4.7K
C10
10u
GND


H1
MountingHole_Pad
H2
MountingHole_Pad
H3
MountingHole_Pad
H4
MountingHole_Pad
GND